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Student Activity Pack

# Why Is Clean Water Becoming Scarce?

Modeling Global Freshwater Systems, Aquifer Depletion, and Water Stress

Grade 11L2 | NGSS HS-ESS2-5, HS-ESS3-1 | Cause and Effect / Systems and System Models

STUDENT NAME _____ DATE _____

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# Table of Contents

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1 Pre-Assessment ..... 3

2 Scoring Rubric ..... 4

3 Component Analysis ..... 5

4 Model Recording Page ..... 6

5 Simulation Observations ..... 7

6 Research & Extend ..... 8

7 Reflection Questions ..... 9

8 STEM Challenge ..... 10

# 1 Pre-Assessment

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**What Do You Know?** Answer each question below to the best of your ability. There are no wrong answers — this helps your teacher understand what you already know!

**1 If Earth is 70% water, why do you think water scarcity exists?**

**2 Where does the water come from that irrigates the food you eat?**

**3 Draw a diagram showing how you think water moves underground and how wells access it.**

**4 What do you think happens to farming communities when the groundwater runs out?**

## 2 Scoring Rubric

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Your teacher will use this rubric to assess your work. Read through each level to understand what is expected!

| Skill | Expert (4) | Proficient (3) | Developing (2) | Beginning (1) |
|---|---|---|---|---|
| **Identifying Components** | EXPERT<br>Correctly identifies and sorts ALL components as internal or external with detailed reasoning | PROFICIENT<br>Correctly identifies and sorts MOST components with some reasoning | DEVELOPING<br>Identifies components but has difficulty sorting or explaining | BEGINNING<br>Needs significant help identifying components |
| **Establishing Relationships** | EXPERT<br>Accurately describes ALL positive and negative relationships with clear evidence | PROFICIENT<br>Accurately describes MOST relationships with some evidence | DEVELOPING<br>Identifies some relationships but struggles with direction | BEGINNING<br>Needs help understanding how components connect |
| **Running Simulations** | EXPERT<br>Makes accurate predictions AND explains surprising results using evidence | PROFICIENT<br>Makes reasonable predictions and records clear observations | DEVELOPING<br>Runs simulations but predictions or observations are incomplete | BEGINNING<br>Needs help running simulations or recording data |
| **Explaining Results** | EXPERT<br>Uses multiple pieces of evidence to explain HOW and WHY systems interact | PROFICIENT<br>Uses evidence to explain system interactions | DEVELOPING<br>Gives partial explanations with limited evidence | BEGINNING<br>Needs help connecting evidence to explanations |

# 3 Component Analysis

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### What is a Component?

A **component** is a part of a system that plays a specific role. Components can be **EXTERNAL** (come from outside the system) or **INTERNAL** (happen within the system).

### Sort These Components

- Population and Agricultural Demand
- Groundwater Extraction Rate
- Aquifer Water Level
- Natural Recharge Rate
- Surface Water Availability
- Regional Food Production Capacity

| EXTERNAL | INTERNAL |
|---|---|
| | |

**Explain your reasoning: Why did you sort each component the way you did?**

# 4 Model Recording Page

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Draw your model showing all components and their relationships. Use the relationship key below to label your arrows!

+ **POSITIVE:** When A increases, B also increases

− **NEGATIVE:** When A increases, B decreases

Draw your model here

**? Explain ONE relationship in your model. What happens when one component changes?**

# 5 Simulation Observations

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Run each scenario in ModelIt! First, write your **prediction**. Then run the simulation and record what you **observe**.

## 1 Scenario 1: Sustainable Withdrawal

Extraction equals recharge, steady state

MY PREDICTION

WHAT I OBSERVED

## 2 Scenario 2: Current Overdraft

Extraction at 3x recharge, declining aquifer

MY PREDICTION

WHAT I OBSERVED

## 3 Scenario 3: Climate + Population

Higher demand, lower recharge, accelerating stress

MY PREDICTION

WHAT I OBSERVED

# 6 Research & Extend

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Real systems are more complex than our model! Here are additional components scientists study. Choose one and explain how it could connect to our model.

## Possible New Components

**Desalination Capacity** — The ability to convert seawater into freshwater using reverse osmosis or thermal distillation — currently supplies less than 1% of global freshwater at costs of $0.50-2.00 per cubic meter, with significant energy requirements and brine waste

**Climate Change Impact on Precipitation** — Shifting rainfall patterns that alter both the total amount and seasonal timing of precipitation, with many agricultural regions projected to receive less rainfall and more extreme droughts while others face increased flooding

**Water Pricing Policy** — The economic mechanism for allocating scarce water — in most regions, agricultural water is heavily subsidized or unpriced, removing the market signal that would incentivize conservation and efficient use

## My Extension

**A Which new component would you add to the model?**

**B How would it connect to the existing components? (Draw or describe)**

**C What evidence or reasoning supports your connection?**

# 7 Reflection Questions

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**The Big Question**

**If Earth is 70% water, why are billions of people running out of clean water — and why can't we just make more?** Use evidence from your simulation to support your answers.

1. **Why does your model show that aquifer depletion is functionally irreversible even though the water cycle continues to operate?**

2. **How does the concept of 'virtual water' change your understanding of water scarcity when applied to global food trade?**

3. **What does your model reveal about why technological solutions alone (desalination, recycling) cannot solve the global water crisis?**

4. **How do feedback loops in your model create a situation where water scarcity makes food scarcity worse, which makes water scarcity worse?**

5. **What does your model suggest about the equity dimensions of water scarcity — who runs out first and why?**

# 8 STEM Challenge: Design a Regional Water Security Plan

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## YOUR ENGINEERING MISSION

Design a comprehensive 30-year water management plan for an agricultural region facing aquifer depletion that maintains food production while transitioning to sustainable water use.

### THE SCENARIO

A major agricultural region producing 15% of the nation's food relies on a depleting aquifer for 80% of its irrigation water. At current withdrawal rates, the aquifer will be unable to support irrigation within 30 years. Your team must design a transition plan that maintains food production while reducing groundwater withdrawal to sustainable levels.

### Engineering Design Thinking

1. WHAT MIX OF EFFICIENCY IMPROVEMENTS, CROP CHANGES, AND ALTERNATIVE WATER SOURCES WOULD CLOSE THE GAP BETWEEN WITHDRAWAL AND RECHARGE?

2. HOW WOULD YOU DESIGN A GROUNDWATER MONITORING NETWORK TO TRACK AQUIFER LEVELS AND ENFORCE WITHDRAWAL LIMITS?

3. WHAT ECONOMIC TRANSITION SUPPORT WOULD FARMERS NEED TO SHIFT FROM WATER-INTENSIVE TO WATER-EFFICIENT CROPS?

4. HOW WOULD YOU PHASE YOUR PLAN OVER 30 YEARS SO THAT FOOD PRODUCTION DISRUPTION IS MINIMIZED DURING THE TRANSITION?

### My Design

DESIGN AREA — DRAW YOUR SOLUTION

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**! Explain your design using scientific evidence from the simulation:**

### Real Career Connection

Hydrologists and Water Resource Engineers study the distribution, movement, and quality of freshwater systems, designing sustainable management strategies for communities, agriculture, and industry. They work for the U.S. Geological Survey, state water agencies, engineering firms, and international development organizations, earning $70,000–$135,000/year. Agricultural Water Management Specialists earn $65,000–$120,000/year.